Project Design Phase – II

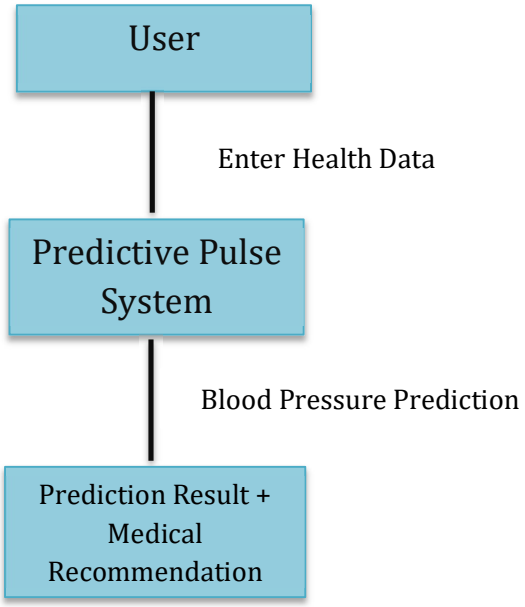
Project Name: Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis

Date: 8 March 2026

Team ID: _____

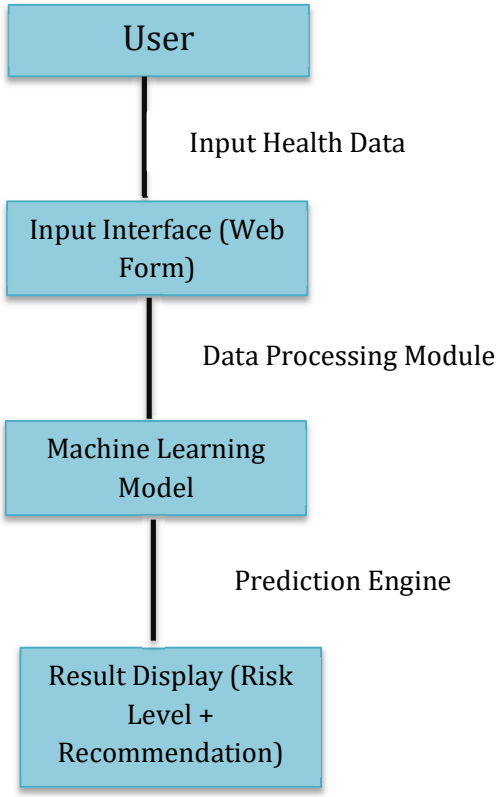
1. Data Flow Diagram (DFD)

Level 0 DFD (Context Diagram)



The user enters health information such as age, systolic pressure, diastolic pressure, and symptoms. The Predictive Pulse system processes the data using a machine learning model and returns the predicted blood pressure stage along with health recommendations.

Level 1 DFD



In Level 1 DFD, the user submits health data through a web interface. The system processes the input and passes it to the trained machine learning model. The model predicts the hypertension stage and the system displays the result and recommendations.

2. User Stories

User Type	Functional Requirement	User Story No.	User Story	Acceptance Criteria	Priority	Release
Patient	Health Data Input	USN-1	User enters health data to check BP risk	Form accepts and submits health data	High	Sprint-1
Patient	Risk Prediction	USN-2	User receives hypertension risk prediction	System displays predicted BP stage	High	Sprint-1
Patient	Medical Recommendation	USN-3	User receives health advice	System shows recommendations	High	Sprint-1
Patient	Result Visualization	USN-4	User views risk level indicators	Results shown with color levels	Medium	Sprint-2
Doctor	Patient Analysis	USN-5	Doctor analyzes patient predictions	Doctor can view prediction results	Medium	Sprint-2
Administrator	System Monitoring	USN-6	Admin monitors system	Admin dashboard shows system status	Low	Sprint-3

3. DFD Explanation

The Data Flow Diagram (DFD) shows how data moves through the Predictive Pulse system. The user provides health data using a web form. The backend processes this information and sends it to the machine learning model. The model predicts the blood pressure risk level such as Normal, Stage 1 Hypertension, or Stage 2 Hypertension. The system then displays the results along with helpful medical recommendations.